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BAGONG PILIPINAS

CC 101 – Introduction to Computing

Laboratory Worksheet No. 3 | Week 6: Computer Networks & the Internet (CO4)

"Know Your Network Settings": Exploring Windows 11 Network Configuration Tools

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Computer Unit Used (Desktop/Laptop, brand): **Date:** 10/18/26

I. Objectives

At the end of this activity, you should be able to:

- Locate network adapter information using Windows 11 Settings, the Control Panel, Command Prompt (CMD), and PowerShell.
- Use **ipconfig**, **getmac**, **hostname**, **ping**, **netstat**, **route print**, and **systeminfo** in CMD, and **Get-NetAdapter**, **Get-NetIPAddress**, and **Test-Connection** in PowerShell.
- Interpret a sample **ipconfig /all** output from a networked computer: identify its IP address, subnet mask, default gateway, and DNS servers, and classify the address as public or private.
- Relate each tool and finding to the layer of the OSI model where it works.
- Design a simple network and IP address plan for the computer laboratory.

II. Materials Needed

- A computer running Windows 11 (a network or Internet connection is **not** required for this activity)
- This worksheet (printed copy, or the digital file to fill out and submit)
- Week 6 lecture notes, for reference on IP addresses, protocols, the OSI model, and network devices

III. Background

A computer does not need to be connected to a network to have network settings. Every PC has one or more **network adapters** (NICs), each with a permanent **MAC address**, and Windows always runs the TCP/IP software, including the **loopback** address 127.0.0.1 (IPv4) and ::1 (IPv6) that the computer uses to talk to itself. Windows 11 lets you inspect these settings in four places: the **Settings** app, the classic **Control Panel**, **Command Prompt (CMD)**, and **PowerShell**. In this activity, you will explore all four on your own unit, observe how Windows behaves when no network is available, and then analyze a sample configuration taken from a computer that *is* connected to a network.

IV. Procedure

Reminder: This is a *viewing* activity. Do not change, disable, or reset any network setting. Close every dialog with **Cancel**. Record values exactly as Windows shows them.

Part A – Windows 11 Settings

1. Press **Windows + I** to open Settings, then click **Network & internet**. Write down the connection status shown at the top of the page (for example, "Not connected").
2. Click **Advanced network settings**. Under **Network adapters**, note every adapter listed (for example, Ethernet, Wi-Fi, Bluetooth Network Connection) and whether each one is enabled

or disconnected.

3. Scroll down and click **Hardware and connection properties**. For each adapter, record its **Name**, **Description**, **Physical address (MAC)**, and **Status** in **Table 1**.

Part B – Control Panel (Network Connections)

4. Press **Windows + R**, type **ncpa.cpl**, and press **Enter**. This opens the classic **Network Connections** window. A red X on an icon means the adapter has no network connection.
5. Right-click your **Ethernet** adapter and choose **Properties**. List the items checked under "This connection uses the following items" in **Table 2**.
6. Select **Internet Protocol Version 4 (TCP/IPv4)** and click **Properties**. Record whether the computer is set to **Obtain an IP address automatically** (DHCP) or **Use the following IP address** (static), and the same for DNS. Click **Cancel**, then **Cancel** again. Do not change anything.

Part C – Command Prompt (CMD)

7. Press the **Windows key**, type **cmd**, and press **Enter**. Run each command below, one at a time, and record the results in **Table 3**.

   hostname

   getmac /v

   ipconfig /all

   Tip: For a disconnected adapter, **ipconfig /all** shows "Media disconnected" and no IPv4 address, but it still shows the Description and Physical Address.

8. Test the computer's own TCP/IP software using the loopback address. Record whether you got replies and the time. For **ping localhost**, note the address that replied.

   ping 127.0.0.1

   ping localhost

9. Now observe what happens when there is no network. Record the **exact** message each command displays.

   ping 8.8.8.8

   nslookup google.com

10. View the ports your computer is listening on and its routing table. In Table 3, record three LISTENING local addresses with their port numbers, and the Netmask shown for the Network Destination 127.0.0.0.

    netstat -an | findstr LISTENING

    route print -4

11. Run the command below. It may take a few seconds. Scroll to **Network Card(s)** and record how many network cards are installed and the status of each.

    systeminfo

Part D – PowerShell

12. Right-click the **Start** button and choose **Terminal** (or press the **Windows key**, type **PowerShell**, and press **Enter**). Run each command below and record the results in **Table 4**.

    Get-NetAdapter

    Get-NetIPAddress -AddressFamily IPv4 | Format-Table InterfaceAlias, IPAddress, PrefixLength

    Test-Connection 127.0.0.1 -Count 4

    Get-NetTCPConnection -State Listen | Select-Object -First 5 LocalAddress, LocalPort

Tip: PowerShell commands follow a Verb-Noun pattern (Get-NetAdapter = "get the network adapters"). Compare each result with the matching CMD command from Part C.

Part E – Analyze a Networked Computer

13. The output below was taken from a laboratory computer that **is** connected to a network. Study it, then complete **Table 5**.

Sample Output 1. ipconfig /all (excerpt) from a networked PC

```
Host Name . . . :                 CICT-LAB-PC07
Ethernet adapter Ethernet:
Description . . . :               Realtek PCIe GbE Family Controller
Physical Address . . . :          3C-52-82-A1-7F-09
DHCP Enabled . . . :              Yes
Link-local IPv6 Address . . . :   fe80::8c2d:4a1f:93b2:61e5%12(Preferred)
IPv4 Address . . . :              192.168.10.37(Preferred)
Subnet Mask . . . :               255.255.255.0
Lease Obtained . . . :            Wednesday, October 14, 2026 8:02:15 AM
Lease Expires . . . :             Thursday, October 15, 2026 8:02:15 AM
Default Gateway . . . :           192.168.10.1
DHCP Server . . . :               192.168.10.1
DNS Servers . . . :               192.168.10.1, 8.8.8.8
```

Part F – Design a Network for Our Laboratory

14. Our laboratory does not yet have a network. In the box after Table 5, draw a simple **star topology** network for it. Include at least **four PCs**, one **switch**, one **router** (default gateway), one **printer**, and the **ISP / Internet**. Use lines to show connections.

15. Assign each device a private IPv4 address from the network **192.168.10.0/24**, using **192.168.10.1** for the router. Label each device in your drawing and complete the **IP Address Plan** (Table 6).

Part G – Optional / Bonus

16. In CMD, run **type C:\Windows\System32\drivers\etc\hosts** to *view* (not edit) the hosts file. In one or two sentences, explain what this file is for and how it relates to DNS.

V. Data and Findings

Table 1. Network Adapters in Settings (Part A)

Connection status shown on the Network & internet page: _____

Adapter Name	Description	Physical Address (MAC)	Status
Ethernet 2	Realtek PCIe GbE Family Controller #2	b0:41:6f:12:c3:4d	Not operational
Ethernet 3	VirtualBox Hot-Only Ethernet Adapter	0a:00:27:00:00:05	Operational
Local Area Connection* 3	Microsoft Wi-Fi Direct Virtual Adapter #3	e4:c7:67:ef:68:e4	Not operational
Local Area Connection* 12	Microsoft Wi-Fi Direct Virtual Adapter #4	e6:c7:67:ef:68:e3	Not operational

Table 2. Adapter Properties (Part B)

Items Checked Under "This connection uses the following items"	**Item Checked (continued)**
Client for Microsoft Networks	✅
File and Printer Sharing for Microsoft Networks	✅
VirtualBox NDIS6 Bridged Networking Driver	✅
QoS Packet Scheduler	✅

TCP/IPv4 Setting	**Automatic or Static?**
IP address	
DNS server address	

Table 3. Command Prompt Results (Part C)

Command / Item	**Your Findings**	**OSI Layer**
hostname	DESKTOP-LGU2FØF	Application
getmac /v: MAC address and Transport Name	Ethernet 2: B0-41-6F-12-C3-4D (Media disconnected) Wi-Fi 2: A2-73-54-A5-C3-B9 Ethernet 3: 0A-00-27-00-00-05	Data Link
ipconfig /all: Media State of Ethernet	Media disconnected	—
ipconfig /all: DHCP Enabled	Yes (Wi-Fi), No (VirtualBox Ethernet 3)	Application
ping 127.0.0.1: replies and time	Packets: Sent=4, Received=4, Lost=0 (0% loss) Approx time: ~0ms	
ping localhost: address that replied	Reply from ::1 — Packets: Sent=4, Received=4, Lost=0 Minimum=0ms, Maximum=0ms, Average=0ms Address that replied: ::1 (IPv6 loopback)	Network
ping 8.8.8.8: exact message	Reply from 8.8.8.8: bytes=32 time=41ms TTL=115 Reply from 8.8.8.8: bytes=32 time=36ms TTL=115 Reply from 8.8.8.8: bytes=32 time=29ms TTL=115 Reply from 8.8.8.8: bytes=32 time=65ms TTL=115	Network

Command / Item	Your Findings	OSI Layer
nslookup google.com: exact message	Server: 192.168.0.1 Address: 192.168.0.1 Non-authoritative answer: Name: google.com Addresses: 2607::4007:80b::200e 142.250.207.46	Application
netstat -an: three LISTENING addresses:ports	TCP Ø.Ø.Ø.Ø:135 Ø.Ø.Ø.Ø:Ø LISTENING TCP Ø.Ø.Ø.Ø: 445 Ø.Ø.Ø.Ø:Ø LISTENING TCP Ø.Ø.Ø.Ø: 33Ø6 Ø.Ø.Ø.Ø:Ø LISTENING	Transport
route print -4: Netmask for 127.0.0.0	255.Ø.Ø.Ø	Network
systeminfo: number of NICs and status	[01] Ethernet 2: Media disconnected [02] Wi-Fi 2: Connected, DHCP: Yes, IP: 192.168.0.126 [03] Ethernet 3: Connected, DHCP: No, IP: 192.168.56.1 —	—

Table 4. PowerShell Results (Part D)

Command	What It Showed	Matches CMD? (Y/N)
Get-NetAdapter: Name, Status, MacAddress, LinkSpeed	Ethernet 2: Disconnected, B0-41-6F-12-C3-4D Wi-Fi 2: Up, A2-73-54-A5-C3-B9 Ethernet 3: Up, 0A-00-27-00-00-05	Y
Get-NetIPAddress: IPv4 addresses and PrefixLength	Wi-Fi 2: 192.168.0.126 /24 Ethernet 3: 192.168.56.1 /24	Y
Test-Connection 127.0.0.1: result	Reply from 127.0.0.1, 4/4 replies, ~0ms	Y
Get-NetTCPConnection -State Listen: two LocalAddress:LocalPort pairs	0.0.0.0:135 0.0.0.0:445	Y

Table 5. Sample Output Analysis (Part E)

Question	Your Answer
IPv4 address of PC07	192.168.10.37
Subnet mask (and its /prefix)	255.255.255.0 (/24)
Default gateway, and what device it usually is	192.168.10.1 — Typically a router
DNS servers	192.168.10.1 and 8.8.8.8
Was the address assigned by DHCP or set manually? How do you know?	DHCP — Lease timestamps shown

Question	Your Answer
Public or private? Which RFC 1918 range?	Private — 192.168.10.0/24 (RFC 1918)
IPv4 address of PC07 in binary (four octets)	11000000.10101000.00001010.00100101
MAC address, and the OSI layer where it is used	3C-52-82-A1-7F-09 — Data Link Layer
How long is the DHCP lease?	24 hours
If the DHCP server failed, what kind of address would PC07 assign itself?	APIPA — 169.254.x.x

Figure 1. Proposed Laboratory Network Diagram (Part F)

Draw or paste your diagram here.

Table 6. IP Address Plan (Part F)

Device	IPv4 Address	Subnet Mask	Default Gateway
Router (gateway)	192.168.10.1	255.255.255.0	—
PC 1	192.168.10.10	255.255.255.0	192.168.10.1
PC 2	192.168.10.11	255.255.255.0	192.168.10.1
PC 3	192.168.10.12	255.255.255.0	192.168.10.1
PC 4	192.168.10.13	255.255.255.0	192.168.10.1
Printer	192.168.10.20	255.255.255.0	192.168.10.1

Part G Notes (Optional / Bonus)

The hosts file is just a simple file on your computer where you can put a website name and its IP address. Your computer checks this file first before asking the internet's DNS server. So if you put something there, your computer uses that instead of looking it up online.

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Even without a network connection, what network information did your computer still have? Why do you think it still has it?

My computer still knew its own name, its MAC address, and the loopback address 127.0.0.1. I think it keeps this info because it's part of the computer itself it doesn't need the internet to know who it is. The loopback address is basically just the computer talking to itself, so it works even when you're offline.

2. Compare Settings, Control Panel, CMD, and PowerShell. Which tool did you find most useful for viewing network information, and why?

For me, Command Prompt was the easiest and most useful. When I typed ipconfig /all , everything I needed showed up right away in one window my IP address, MAC address, gateway, and all that. I didn't have to click through different menus. The other tools are okay too, but Command Prompt was just faster and simpler for this activity.

3. What did the messages from ping 8.8.8.8 and nslookup google.com tell you? How would you use messages like these when troubleshooting a real network?

Ping basically asks, "Can I reach this place and how fast does it reply?" It checks if you're actually connected to the internet. Nslookup asks, "Do I know where google.com is?" It checks if your DNS is working. So if ping works but nslookup doesn't your internet is fine, but your DNS is broken. If neither works your internet connection is probably down. Using both helps me figure out exactly what's wrong.

4. Why should you be careful about sharing your MAC address, IP address, or ipconfig /all screenshots online? Relate your answer to data privacy.

Because these are like your computer's personal identifiers. Your IP can show roughly where you are and who your internet provider is. Your MAC address is like a unique serial number for your device no other device in the world has the same one. If people get these, they could track you or even try to mess with your connection. It's just like not sharing your personal info keep it private to stay safe online.

VII. Scoring Guide

Criteria	Description	Points
Completeness	All tables (1–6) and required fields are filled out.	25
Accuracy	Recorded values match what the tools show and are consistent across Settings, CMD, and PowerShell; sample analysis, binary conversion, and OSI layers are correct.	30
Network Design	Diagram shows a star topology with PCs, switch, router, printer, and ISP; IP plan uses valid, non-duplicate addresses in 192.168.10.0/24.	15

Criteria	Description	Points
Realization / Reflection	Answers are thoughtful, based on the student's own results, and well written.	20
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	10
Total		**100**